Gestão de Inovação e Design Thinking
Innovation Management and Design Thinking
2026/27

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Office hours

Wednesday 09h30-11h30

(scheduled in advance)

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Course Plan

- Innovation and Strategy
- Design Thinking Overview
- Inspire
- Ideate
- Implement
- DT and Organization Change
- Pitch

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Bibliography

Designing for Growth: A manager's design thinking toolkit: Jeanne Liedtka and Tim Ogilvie 2011 COLUMBIA UNIVERSITY PRES

Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation: Tim Brown 2009HarperCollins

Democratizing Innovation: Eric von Hippel 2055 MIT Press

Goffin, K. and Mitchell, R. (2017) Innovation Management: Effective Strategy and Implementation (3rd edition). Palgrave.

Schillings, M. (2019) Strategic Management of Technological Innovation (6th edition). McGraw-Hill Education.

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Course Evaluation

Group Evaluation (70% of the grade) –

Design Thinking Project

Week 8 - Nov 4th and Nov 5th	Mid term presentation – in class (25%)
16th and 17th December	Final Pitch Presentation (25%)
21 December	Project report (50%)

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Course Evaluation

Individual Evaluation (30% of the grade)

Five in- class task submissions and class participation (20%)

- 22nd September
- 7th October
- 21st October
- 18th November
- 25th November

4 submissions mandatory (5= bonus)

Peer evaluation (10%)

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Classes Methods



Introduction of concepts



Tools & Application

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What is innovation?



What is Innovation?

“ An innovation in the economic sense is accompanied with the first commercial transaction involving the new product, process, system or device, although the word is used to describe the whole process.” Christopher Freeman (1982)

Process

Discovery, experimentation, development and adoption of novelties

Origin

From the Latin *innovare*: to renew, to make new, to change

R&D

Creative work to increase knowledge and develop applications



Innovation vs. Invention





Invention

- New idea or model
- Creation of a device
- Conversion of money into ideas

Innovation

- Commercial transaction
- Introduction of change
- Conversion of ideas into money

Types of Innovation

Categories of innovation according to Joseph Schumpeter (1934):

New Products

Introduction of new goods to the market

New Processes

Introduction of new methods of production

New Markets

Expansion into unexplored areas

New Sources

Different raw materials and inputs

New Structures

Reorganisation of industries





Marketing Innovation

New Methods

Novel approaches to promote products

Design and Packaging

Changes in product presentation

Sales Channels

Introduction of new distribution forms



Organisational Innovation

New Internal Practices

Reorganisation of business routines and procedures

Redistribution of Responsibilities

New methods for assigning functions and decisions

External Relationships

Innovative collaborations with suppliers and partners

(Oslo Manual, OCDE 2005)

(Oslo Manual, OCDE 2005)



Nature of Innovation

Radical Innovation

New technology, new market

Architectural Innovation

Existing technology, new market

Disruptive Innovation

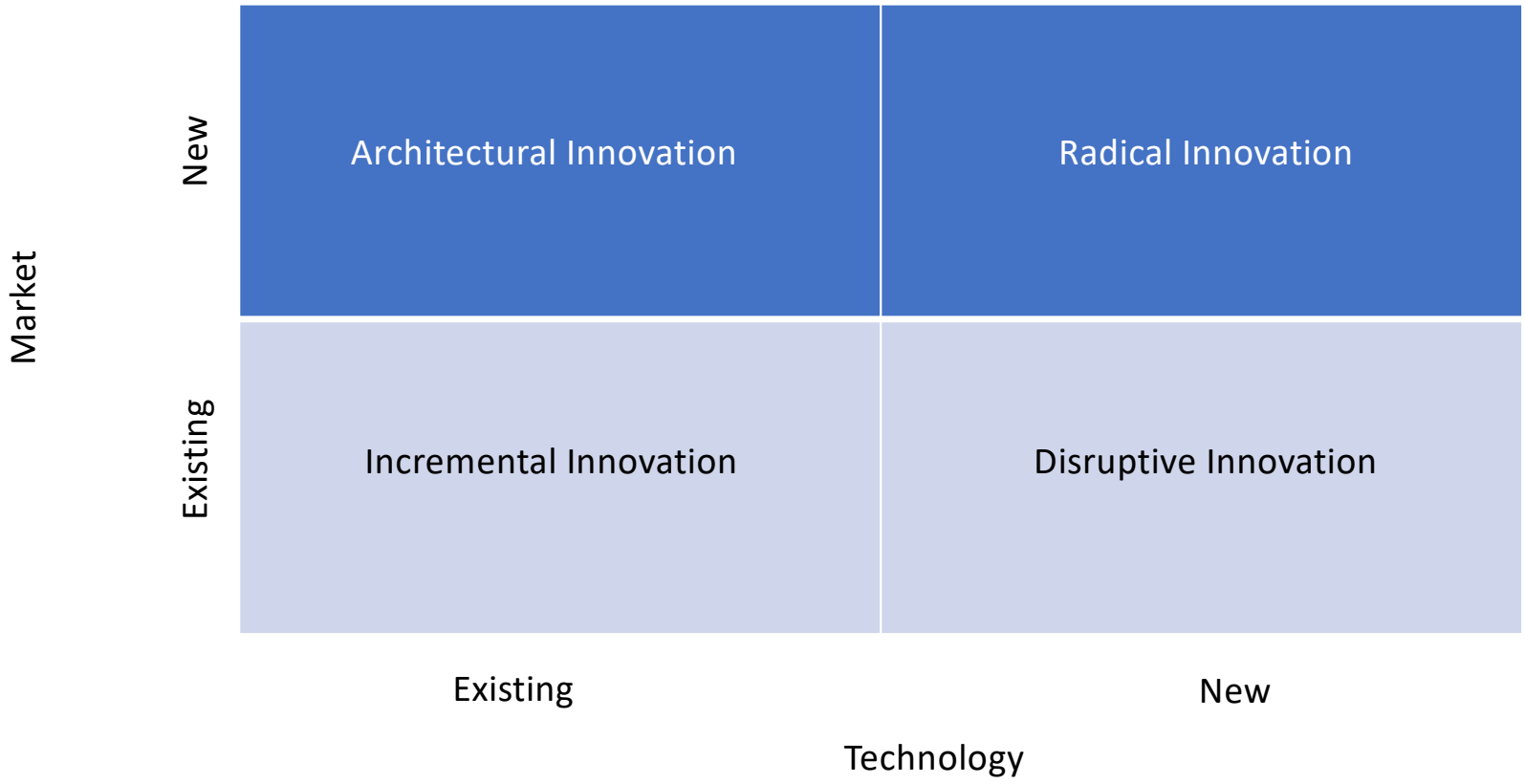
New technology, existing market

Incremental Innovation

Existing technology, existing market

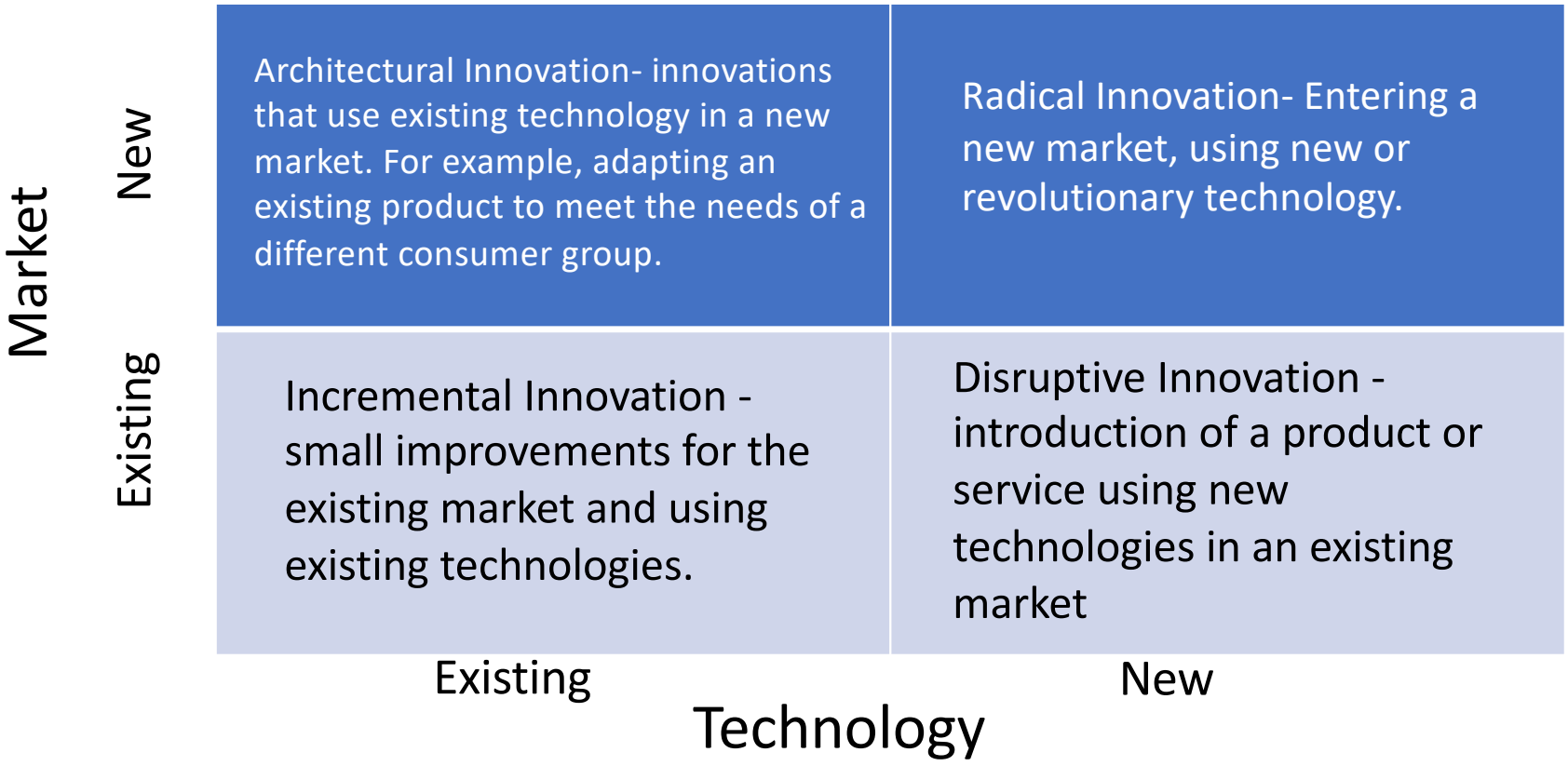
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Types of Innovation



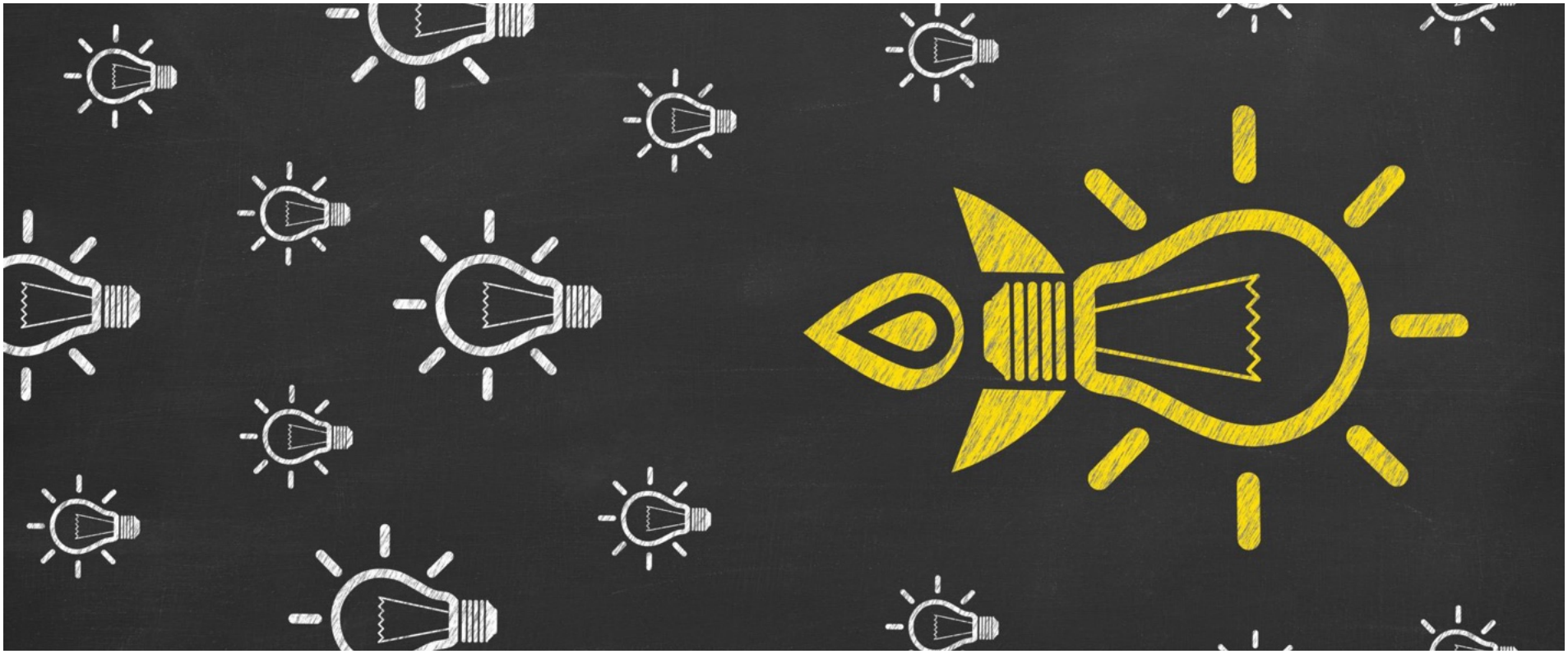
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Types of Innovation



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Where does Innovation come from?



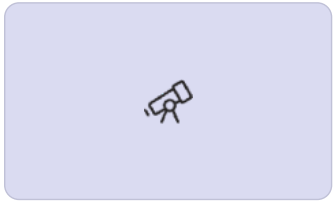
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Technology-Push ou *Capability-Push*

Technology-push: technological advances driven mainly by technical or scientific discoveries, only secondarily orientated to the market specific needs

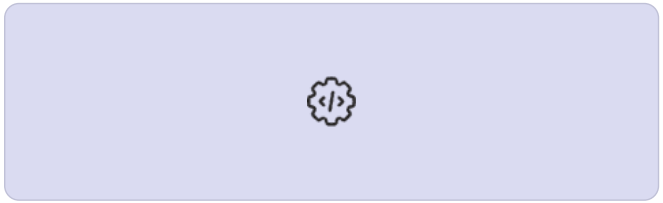
- Innovation oriented primarily towards finding new applications for technical improvements in products and processes
- Technical knowledge concentrated in those responsible for innovation
- They are the most frequent source of radical changes in markets and companies' organizational structures
- Examples: steam machines; penincillin and antibiotics; transistors and microprocessors; PCs; digital photography, cinema and television; biosensors

Technology-Push Innovation



Scientific Discovery

Technical breakthroughs occur in research settings.



Application Development

Potential commercial uses for new technology are explored.



Market Identification

Target customers for the new technology are identified.

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Market-Pull or *Opportunity-pull*

Market pull: technology advances made in the sequence of identified market need, and only secondarily motivated by discoveries of technical or scientific nature

- Innovation aimed primarily at satisfying the needs of a specific market or consumer group
- More often incremental in nature, associated with improvements to technologies already available on the market - the perception of opportunity based on markets and products
- Examples: drugs that affect the chemical composition of the human body specifically aimed at treating nervous depression, or Alzheimer's disease; development of new sail designs for Windsurfing

Market-Pull Innovation

Market Need Identified

Research reveals specific customer problems requiring solutions.

Solution Development

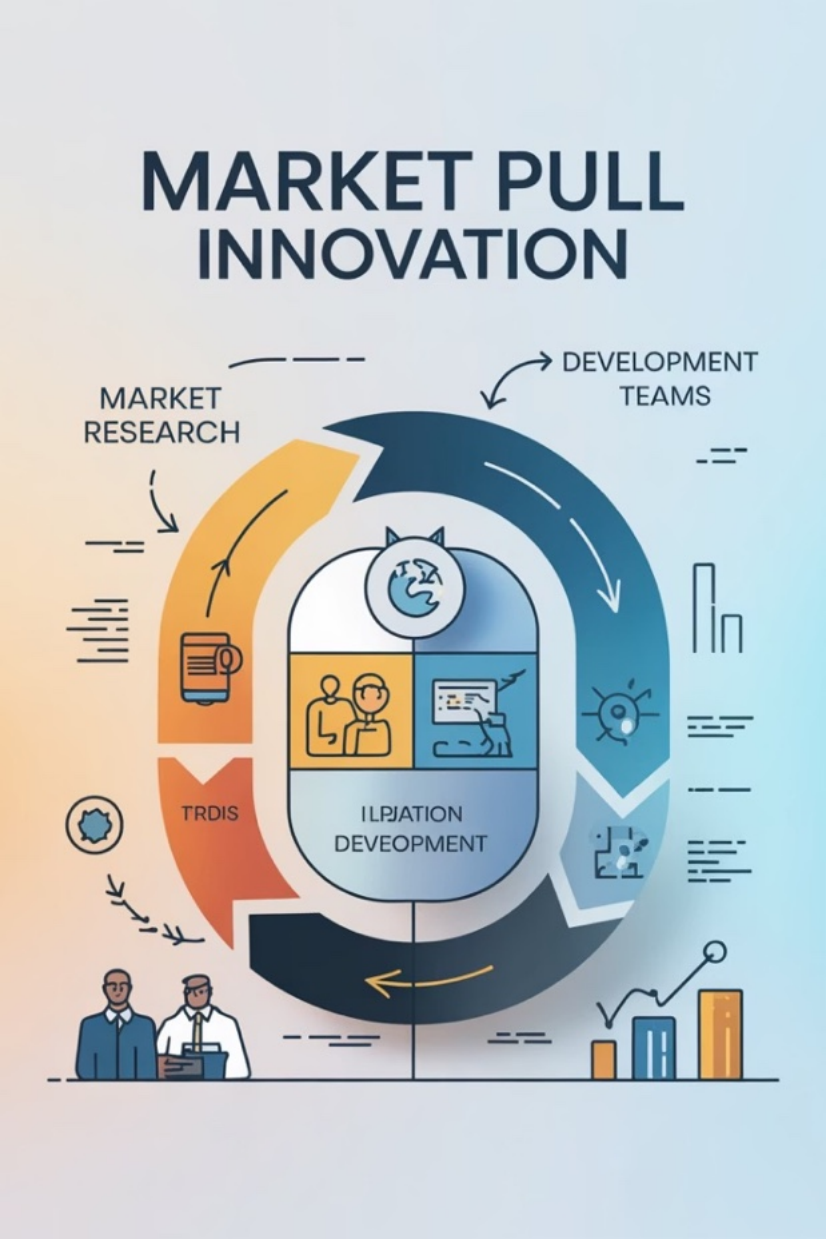
Technical teams work to address the identified need.

Solution Validation

Products tested against original market requirements.

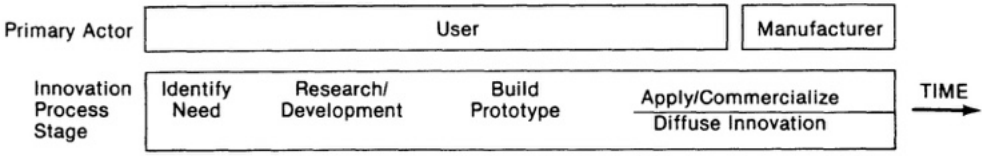
Market Launch

Solution introduced to satisfy the initial need.



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User Innovation



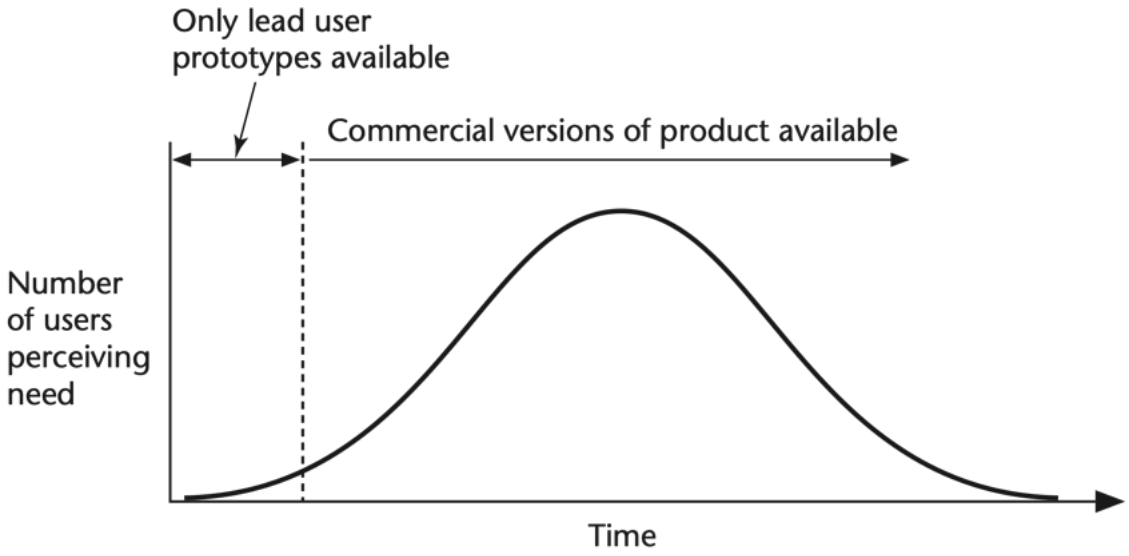
"series of studies showing that the sources of innovation vary greatly... test some implications of replacing the manufacturer-as-innovator assumption with the view of the innovation process as predictably distributed across users, manufacturers, suppliers and others."

Eric von Hippel, *The Sources of Innovation*, Oxford University Press (1988)

- Users, firms or individuals, are important for the innovation process, and have an active role in the development of new products and services
- Consumers have experience of use and participate directly in the development of the new product
- Users create or modify products and services in order to fulfil their own needs, and it is not uncommon that these innovations are made freely available to other users and even to producers

(Harhoff et al., 2003; Oliveira and von Hippel, 2010)

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User Innovation

- user innovation suggests an alternative to mass customization- manufacturing process that seeks to tailor products to specific users while maintaining the economies of large-scale production

Source: Eric von Hippel, Democratizing Innovation,2005



User Innovation: Mountain Bike

User Need Identification

Gary Fisher identified the need for off-road biking equipment in 1979.

User-Led Development

Fisher created his own solution rather than waiting for manufacturers.

Market Transformation

User innovation led to an entirely new product category.

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World Wide Web invented by Tim Berners-Lee (1989)

Created to share data with physicists globally



User-driven solutions often lead to breakthrough innovations

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User Innovation

- ”(…)I grabbed one of the Crocs, pulled some buttons, rhinestones and fabric out of the sewing kit, and stuck them in the holes. Lexie said, 'Mum, I love that!’”
- Sheri and Lexie, then seven, spent the rest of the day filling holes in the family's 12 pairs of Crocs. Every look-alike shoe was suddenly unique.



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Technology

"Technology" includes all equipment, artifacts, processes, tools, methods and materials that can be applied in the production and marketing of products and services

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Apple I (1975)





Xerox Alto (1972)

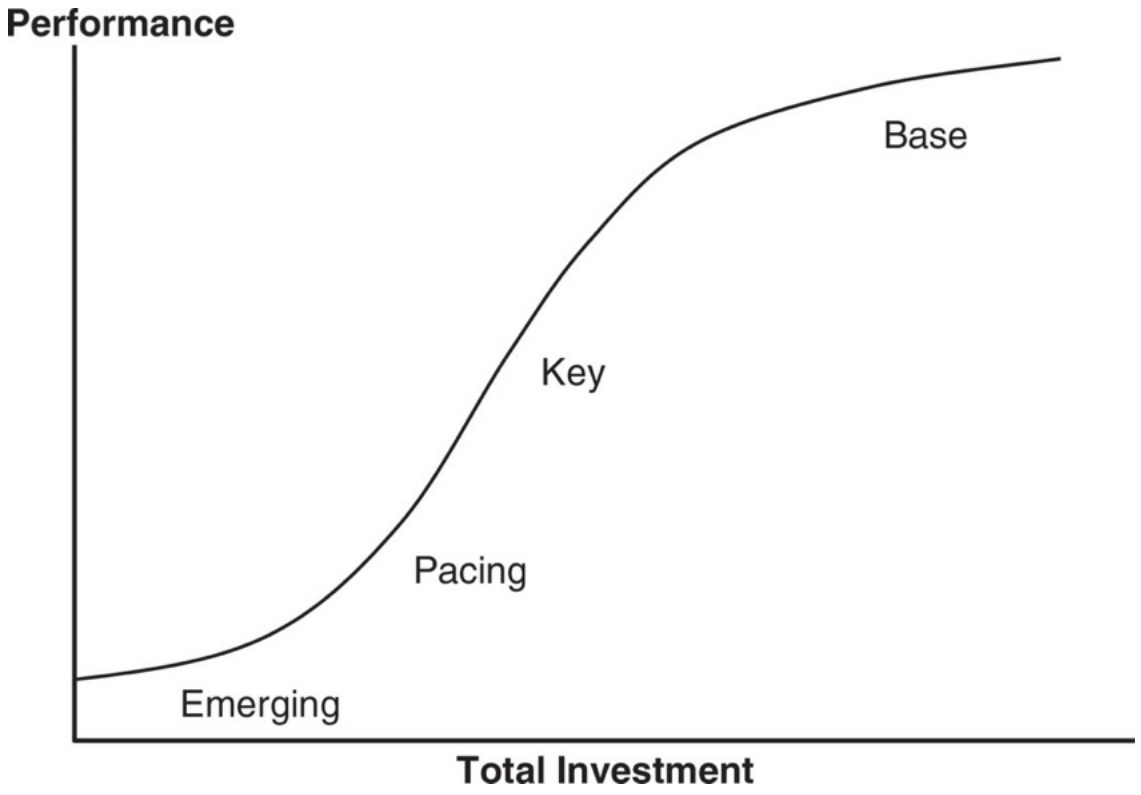
Apple II (1977)





IBM PC (1981)

Technology S-curve



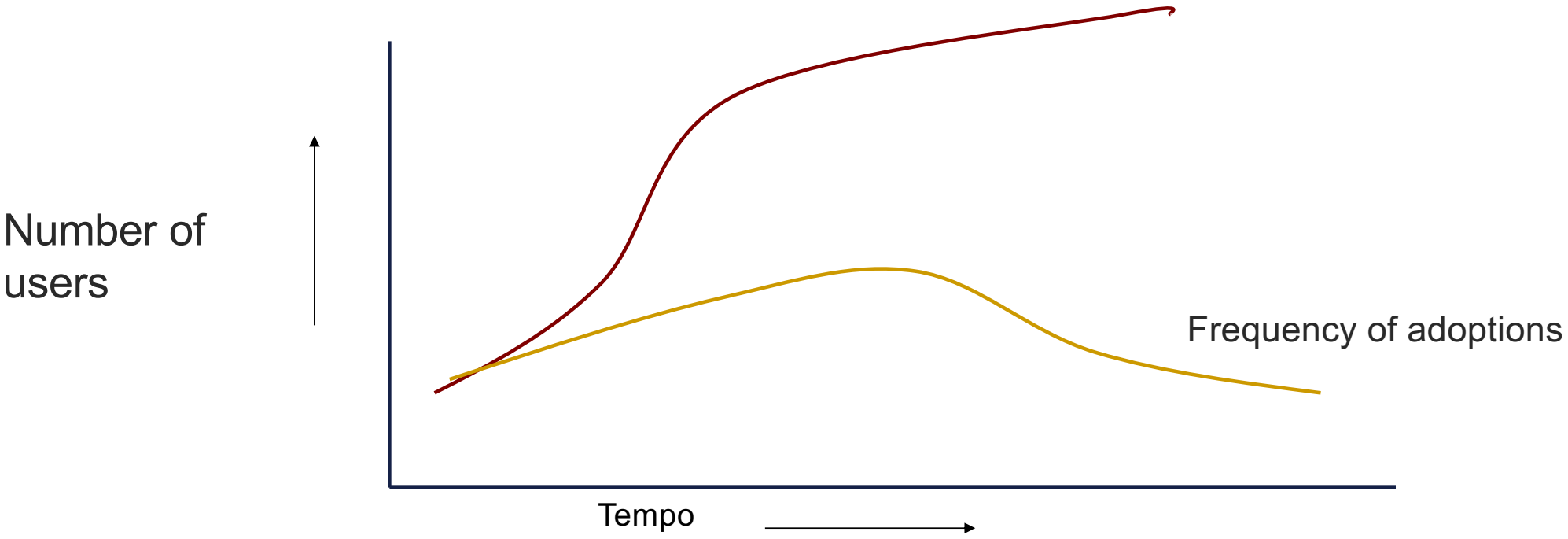
Source: based on Foster, 1996

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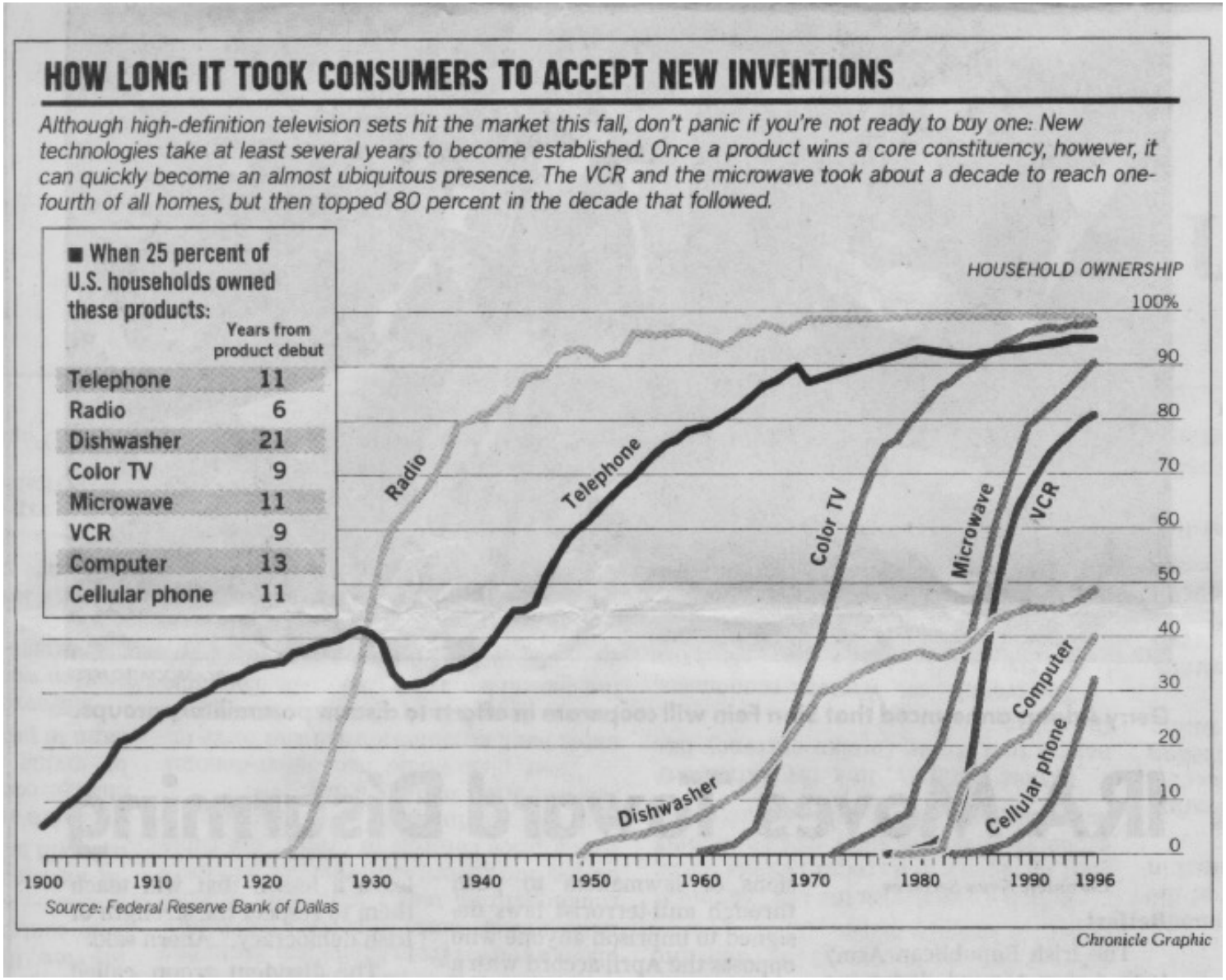
Innovation Diffusion

<u>Diffusion</u> is the process by which an innovation is spread accross time

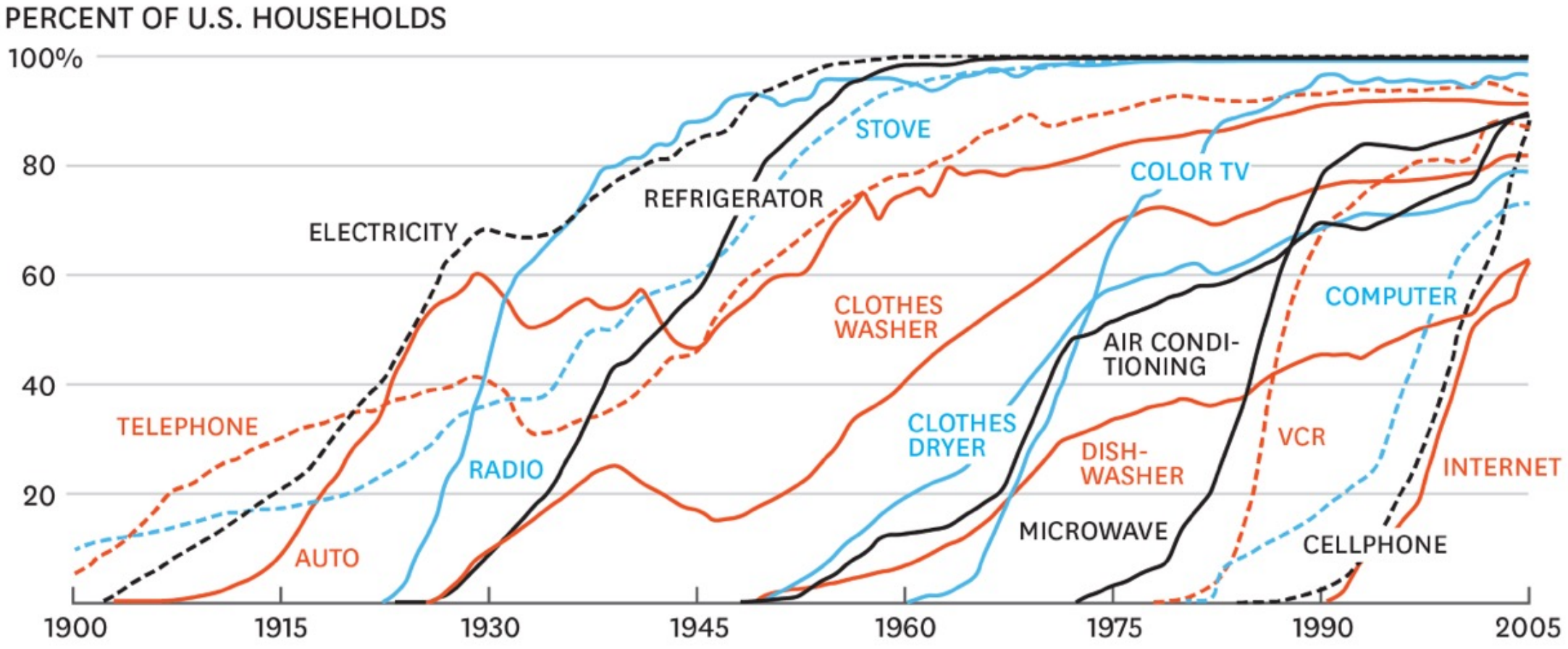
"S" curve of Diffusion



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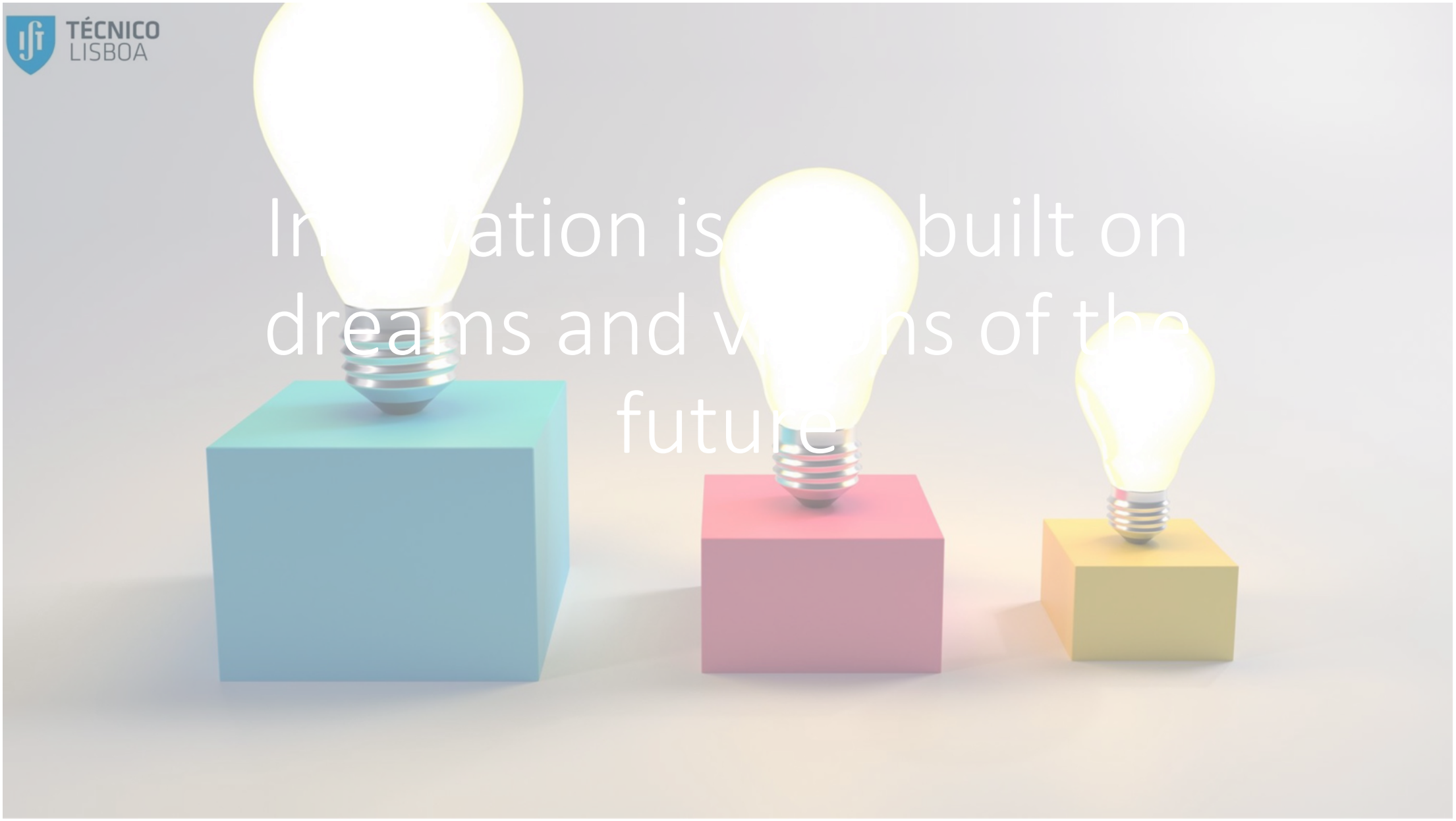


CONSUMPTION SPREADS FASTER TODAY



SOURCE NICHOLAS FELTON, THE NEW YORK TIMES

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Innovation is built on
dreams and visions of the
future